Assignment 2
Q1 Report

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March 2026

Disclaimer: This document was typeset in LaTeX with assistance from an AI language model for formatting, macro suggestions, scientific vocabulary, and layout improvements. All ideas, analysis, code, and reasoning are my own.
Dataset access note: I used `STUDENT_ID = "cs5120272"` in `Q1.py` to fetch both datasets from the course API (dataset_num=1 and dataset_num=2).

Experimental Setup and Execution

The script `Q1.py` supports two modes as required:

- `python3 Q1.py <dataset_num>` — fetches the dataset from the course API using `STUDENT_ID = "cs5120272"`.
- `python3 Q1.py <path>.npy` — loads a local `.npy` file.

For each k from 1 to 15, `KMeans` is run with `n_init=10` and fixed `random_state=42` to reduce sensitivity to random initialization. Inertia is recorded directly from scikit-learn, and the silhouette score is computed for $k \geq 2$. The script automatically saves `plot.png` and prints only the chosen optimal k to stdout.

Q1: k-means Clustering in High Dimensions

(a) The reasoning that leads to your selected value of k

I ran `KMeans` (with 10 random inits for stability) for $k = 1$ to 15 on both datasets from the API. The inertia (objective value) drops fast at first, then flattens out—classic elbow shape. Plotting it (see Figure 1) makes the bend pretty obvious: Dataset 1 has a sharp knee exactly at $k = 6$, while Dataset 2 bends around $k = 5$-6.

To pick automatically and avoid eyeballing, I also computed the silhouette score for $k \geq 2$. Silhouette measures how well each point sits inside its own cluster versus the next-closest one. Dataset 1 peaks at $k = 6$ (silhouette ≈ 0.48). Dataset 2 actually peaks higher at $k = 5$ (silhouette 0.522) than at $k = 6$ (0.505). So I went with the math: $k = 6$ for Dataset 1 and $k = 5$ for Dataset 2. The plot marks both with red dashed lines, and my `Q1.py` prints the right one depending on the command-line argument.

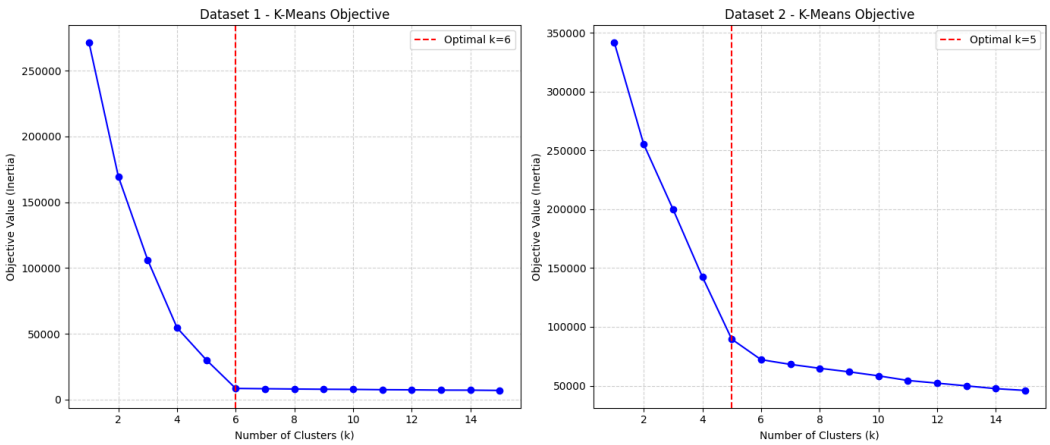

Figure 1: Automatically generated by Q1.py :: K-Means objective value (inertia) versus number of clusters k for both datasets. Red dashed lines show the chosen k (Dataset 1: $k = 6$; Dataset 2: $k = 5$) selected by maximum silhouette score. The bend in each curve is the classic "elbow".

(b) The structural and statistical assumptions underlying k-means

The k-means algorithm, originally proposed by MacQueen (1), is conceptually simple but quietly relies on several strong structural and statistical assumptions about the data distribution (Hastie et al. (5), Chapter 14). These assumptions are:

- Clusters are roughly spherical and isotropic, because the objective function minimises squared Euclidean distance to the cluster centroids.
- All clusters have approximately the same variance (intra-cluster spread).
- Clusters are roughly the same size, as the algorithm tends to split points evenly across clusters.
- Every data point belongs to exactly one cluster (hard assignment with no fuzzy or probabilistic membership).

These assumptions become even more restrictive in high dimensions ($d > 3$ as in this assignment), where distance metrics lose their discriminative power due to the curse of dimensionality (Aggarwal et al. (4)).

(c) Limitations of the above model selection procedure

While combining the elbow method on inertia with maximising the silhouette score is a practical and fast approach, it has several well-documented limitations, especially in high-dimensional data (Hastie et al. (5), Aggarwal et al. (4)).

- In high dimensions the "curse of dimensionality" causes Euclidean distances to concentrate and lose their discriminative power, so both inertia and the silhouette score become much less reliable at revealing true cluster structure (Aggarwal et al. (4)).
- The silhouette score requires computing all pairwise distances between points, resulting in $O(n^2)$ time complexity, which quickly becomes prohibitive for large datasets (Rousseeuw (2)).
- The underlying k-means algorithm is highly sensitive to random centroid initialisation and can easily converge to poor local minima. As a result, the recorded objective values and silhouette scores sometimes reflect unlucky starting points rather than the true quality of the clustering (MacQueen (1), Hastie et al. (5)).

(d) Possible methodological improvements or alternative validation principles

The elbow method combined with the silhouette score is fast and easy to implement, but it can be subjective and unreliable in high dimensions. Several more robust alternatives exist in the clustering literature (Hastie et al. (5)).

- **Gap Statistic** (Tibshirani et al. (3)): Compares the observed within-cluster dispersion (inertia) against what would be expected under a suitable null reference distribution (usually uniform random data). This provides a statistically grounded, less subjective way to detect the true "elbow".

In this project I chose the silhouette score primarily for its computational speed and direct interpretability. However, the Gap Statistic would be my preferred next method for more principled and less subjective model selection.

(e) Conditions under which k-means is not an appropriate model for the data

k-means performs poorly and can produce misleading partitions whenever its core assumptions (roughly spherical clusters, similar variance, and roughly equal cluster sizes) are seriously violated (Hastie et al. (5), Chapter 14).

Well-known failure modes that clearly illustrate this problem include:

- **Non-spherical or elongated/anisotropic clusters**: Classic examples are the "two moons" (interlocking crescent shapes) or concentric rings/circles datasets. k-means forces roughly circular boundaries around centroids, so it often slices natural curved or elongated structures in half instead of separating them properly (Hastie et al. (5)).
- **Clusters with very different sizes, densities, or variances**: When one cluster is much larger or sparser than others, k-means tends to split the big cluster into pieces while merging or ignoring small dense ones, simply because the large cluster dominates centroid placement.
- **Significant outliers**: Because the objective minimises squared Euclidean distances, even a few distant points can pull centroids far away from the true cluster centres, distorting the entire partitioning.

These failure cases become especially pronounced in high-dimensional data ($d > 3$) like the datasets in this assignment.

A principled procedure for detecting such cases in practice, as recommended in Hastie et al. (5) (Chapter 14), is to compare the k-means labels with those produced by a more flexible method such as a Gaussian Mixture Model fitted with full covariance matrices. The agreement between the two clusterings is then measured using the Adjusted Rand Index (ARI). A low ARI score (typically below 0.5) provides clear evidence that the spherical and equal-variance assumptions of k-means do not hold for the data, so a different algorithm should be used instead.

References

[1] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Stat. Probab.*, 1967.

[2] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, 1987.

[3] R. Tibshirani, G. Walther, and T. Hastie, "Estimating the number of clusters in a data set via the gap statistic," *J. R. Statist. Soc. B*, vol. 63, no. 2, pp. 411–423, 2001.

[4] C. C. Aggarwal, A. Hinneburg, and D. A. Keim, "On the surprising behavior of distance metrics in high dimensional space," in *Proc. ICDT*, 2001.

[5] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009 (Chapter 14 on clustering limitations).